

# Tutorial and Practice Questions 1: Splines

## Topics in Advanced Regression

July 2026

### Constrained linear least squares

Minimize  $(Y - X\beta)'(Y - X\beta)$  subject to constraints  $C\beta = b$ , using Lagrange multiplier  $\lambda$ :  $\lambda'(C\beta - b)$

Solution:

$$\begin{bmatrix} 2X'X & C' \\ C & 0 \end{bmatrix} \begin{bmatrix} \beta \\ \lambda \end{bmatrix} = \begin{bmatrix} 2X'Y \\ b \end{bmatrix}$$

Reference: <https://people.duke.edu/~hpgavin/cee251/constrained-least-squares.pdf>

### Question 1

Simulate data from the following model, where  $\mathbf{x}$  is a vector of 100 random values sampled from  $U(0, 1)$ . Define two knots, one at 0.33 and the other at 0.66.

$$y_i \sim N(\mu_i, 0.5^2), \quad \mu_i = 5 + \sin(3\pi(x_i - 0.6))$$

For each of the following define the basis functions / matrix and plot these functions. Then fit them to the data above and create a plot to illustrate the fitted curves, lines.

- Piecewise constant
- Piecewise linear
- Piecewise linear, and continuous at knots (broken stick)
- Piecewise cubic polynomial
- Piecewise cubic polynomial, and continuous at knots
- Piecewise cubic polynomial, and continuous up to 1st derivative at knots
- Piecewise cubic polynomial, and continuous up to 2nd derivative at knots
- Cubic spline (using truncated power basis)

- Natural spline (for the moment use the `ns()` function in package `splines`)
- B-spline basis function (for the moment use the `bs()` function in package `splines`, but see exercise below)

Create another plot where you add confidence bands to one of the fitted splines (natural or B-spline).

Some functions / ideas that you may find useful:

- `cut()` will cut the range of a variable  $x$  into intervals.
- Interactions in linear models are useful for estimating a different line or curve in each section (not continuous).

## Question 2

Write an R function to create a cubic spline basis based on a truncated power basis. Construct the basis matrix  $X$ . Fit a cubic spline to the LIDAR data. Add the fitted spline curve to a plot with the observations. The LIDAR data are available in the `SemiPar` library.

## Question 3

Construct the B-spline basis functions (see ESL Appendix pg. 186): Haar basis, linear, quadratic, cubic B-spline basis functions.  $x$  between -1, 1. Create basis matrix  $X$ . Plot the basis functions.

## Penalized Regression Splines

In practice (e.g. in R packages) regression splines are splines where the user chooses the knots and the spline is fitted with or without penalty (in which case they become penalized regression splines).

We can solve the penalized regression problem as follows:

$$\hat{\beta} = (X'X + \lambda\Omega)^{-1}X'Y$$

The main difficulty here is: how to obtain  $\Omega$ , the penalty matrix? This is one reason for the popularity of P-splines, for which the penalty matrix is easy to calculate (see notes).

Fit P-splines to the LIDAR data. Start with a B-spline basis, calculate the penalty matrix, and fit to the data. Repeat for  $\lambda = 0, 0.1, 0.5, 1, 5, 10$ . Plot the data and add all of the resulting P-splines to the plot.

Repeat this for a range of  $\lambda$  values between 0 and 20, each time calculating the GCV value. Plot GCV as a function of  $\lambda$  and choose a suitable  $\lambda$ .

## Smoothing Splines

In practice (e.g. in R packages) smoothing splines are handled like the regression splines above (basically, a penalized regression spline). A large (enough) number of knots is chosen, and then a roughness penalty is added. The smoothing parameter is often chosen using GCV. So, there is nothing extra you need to do here.

Source: Introduction to Statistical Learning (James et al.), Chapter 7 Exercise 1

A cubic regression spline with one knot at  $\xi$  can be obtained using a basis of the form  $x, x^2, x^3, (x - \xi)_+^3$ , where  $(x - \xi)_+^3 = (x - \xi)^3$  if  $x > \xi$  and equals 0 otherwise. We will now show that a function of the form  $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x - \xi)_+^3$  is indeed a cubic regression spline, regardless of the values of  $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ .

- (a) Find a cubic polynomial  $f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$  such that  $f(x) = f_1(x)$  for all  $x \leq \xi$ . Express  $a_1, b_1, c_1, d_1$  in terms of  $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ .
- (b) Find a cubic polynomial  $f_2(x) = a_2 + b_2 x + c_2 x^2 + d_2 x^3$  such that  $f(x) = f_2(x)$  for all  $x > \xi$ . Express  $a_2, b_2, c_2, d_2$  in terms of  $\beta_0, \beta_1, \beta_2, \beta_3, \beta_4$ . We have now established that  $f(x)$  is a piecewise polynomial.
- (c) Show that  $f_1(\xi) = f_2(\xi)$ . That is,  $f(x)$  is continuous at  $\xi$ .
- (d) Show that  $f'_1(\xi) = f'_2(\xi)$ . That is,  $f'(x)$  is continuous at  $\xi$ .
- (e) Show that  $f''_1(\xi) = f''_2(\xi)$ . That is,  $f''(x)$  is continuous at  $\xi$ .

Therefore,  $f(x)$  is indeed a cubic spline.

Hint: Parts (d) and (e) of this problem require knowledge of single variable calculus. As a reminder, given a cubic polynomial  $f_1(x) = a_1 + b_1 x + c_1 x^2 + d_1 x^3$ , the first derivative takes the form  $f'_1(x) = b_1 + 2c_1 x + 3d_1 x^2$  and the second derivative takes the form  $f''_1(x) = 2c_1 + 6d_1 x$ .

Suppose we fit a curve with basis functions  $b_1(X) = X$ ,  $b_2(X) = (X - 1)^2 I(X \geq 1)$ . (Note that  $I(X \geq 1)$  equals 1 for  $X \geq 1$  and 0 otherwise.) We fit the linear regression model

$$Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + e$$

Sketch the estimated curve between  $X = -2$  and  $X = 2$ . Note the intercepts, slopes, and other relevant information.

Suppose we fit a curve with basis functions  $b_1(X) = I(0 \leq X \leq 2) - (X - 1)I(1 \leq X \leq 2)$ ,  $b_2(X) = (X - 3)I(3 \leq X \leq 4) + I(4 < X \leq 5)$ . We fit the linear regression model

$$Y = \beta_0 + \beta_1 b_1(X) + \beta_2 b_2(X) + e,$$

and obtain coefficient estimates  $\hat{\beta}_0 = 1, \hat{\beta}_1 = 1, \hat{\beta}_2 = 3$ . Sketch the estimated curve between  $X = -2$  and  $X = 2$ . Note the intercepts, slopes, and other relevant information.

## Elements of Statistical Learning (Hastie et al.)

Exercise 5.1 - 5.4

Simulate data:

- two order degree 4, joined at  $a$
- values and derivatives continuous at  $a$

Use two methods to fit the spline. Do they give you the same estimated spline?

## Explore

1. How do the natural spline functions in R work. Do they give you what you expect?
2. Construct B-splines from first principles (recursive equations). Where does this idea come from (convolution?). What does R do? Does this correspond to your splines? Intercept?
3. There are other types of splines (I, M,  $\dots$ ). Explore these.
4. Fit various types of splines using a Bayesian approach (Stan).
5. Reinsch form of cubic splines (ESL), chapter 5
6. Splinets <https://journal.r-project.org/articles/RJ-2024-034/>
7. Variance and model selection for penalized splines.

## Classical Data Sets

1. Wage data in ISLR: income vs age
2. Motorcycle data: `mcycle` in MASS. Acceleration of head of crash test dummy over time.
3. wine
4. CO2 data: `co2` in R. Mauna Loa atmospheric CO2 concentration.
5. Bone density with age
6. Cherry blossom data